

Archit Khandelwal

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SUMMARY

Computer Science undergraduate at VIT Chennai (CGPA 8.86) who builds and ships web and mobile applications end to end: FastAPI backends, React/Next.js frontends, and Flutter apps. Delivered a live production website for a startup client and built ShelfSense, an offline Android app backed by 113 unit tests. Strong in DSA, OOP, and system design; a quick learner across web, mobile, cloud, and data. Seeking a Software Engineering Internship building digital platform solutions.

EDUCATION

B.Tech in Computer Science and Engineering — Vellore Institute of Technology, Chennai Expected May 2028
CGPA: 8.86 / 10

Minor in Software Development 2.0 — Indian Institute of Technology, Mandi Aug 2026
Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), System Design, Cloud Computing & Deployment, Programming Fundamentals, Generative AI & Applications

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, REST API Design, DBMS
Web & Backend: React 18, Next.js 14, FastAPI, Pydantic, Vite, React Router, Tailwind CSS, Radix UI, HTML/CSS, Responsive UI Design
Mobile: Flutter, Dart, Android, TensorFlow Lite (on-device ML), SQLite
Databases: PostgreSQL, MySQL, SQLite, Supabase, Redis (NoSQL key-value store)
Cloud & DevOps: AWS, Docker, Git, GitHub, Linux / Bash
AI / ML & Data: PyTorch, TensorFlow, scikit-learn, Hugging Face (Transformers, PEFT), QLoRA / LoRA, OpenAI Whisper, OpenCV, pandas

EXPERIENCE

Freelance Full-Stack Developer — EduSphere Feb – Mar 2026
React · TypeScript · Tailwind CSS | edusphereofficial.in | github.com/ARCHIT3024/Edusphere

- Gathered requirements directly from a startup client and delivered a production EdTech website that is live and serving real users.
- Built 13 mobile-first pages (down to 320px) with React, TypeScript, Tailwind CSS v4, Radix UI, and React Router v7: a 70+ course catalog with search and filtering, 3-step EmailJS registration, Google Maps + WhatsApp integrations, and SEO-friendly HTML.

Campus Ambassador — Unstop Apr 2025 – May 2026

- Represented Unstop at VIT Chennai for 13 months, promoting hackathons and coding competitions to grow student participation.

PROJECTS

ShelfSense — Offline Mobile Shelf Audit & Ordering App Sep 2026
Flutter · Dart · SQLite · TensorFlow Lite · YOLO11n · MobileNetV3 | github.com/ARCHIT3024/ShelfSense

- Built with a 3-person team in a 2-day hackathon (iQOO City Battle Chennai): a fully offline Android app that photographs a store shelf, identifies every product, compares it against the planned layout, and drafts a reorder exported as XLSX, CSV, and PDF.
- Engineered for reliability: 113 unit tests, zero static-analysis errors, a 14-table SQLite schema, model contract checks that fall back to manual mode instead of crashing, and a local Wi-Fi server with QR code for file handover.
- Integrated a two-stage on-device ML pipeline (YOLO11n INT8 detector with mAP50 0.889 on 2,935 held-out images; MobileNetV3 embedder with cosine KNN matching), returning labelled results ~760 ms after shutter press.

MedScript — AI Clinical Documentation Platform Jul – Aug 2026
Python · FastAPI · Next.js 14 · TypeScript · NVIDIA NIM · OpenAI Whisper · QLoRA | github.com/ARCHIT3024/Medscript

- Designed a FastAPI backend for high availability with dual-path inference (cloud NVIDIA NIM Nemotron-70B as primary, a locally hosted fine-tuned model as fallback) and HL7 FHIR R4 export for Electronic Health Record (EHR) integration.
- Built the end-to-end voice-to-SOAP-note flow with under 3 s latency: Whisper speech-to-text, note generation, editable SOAP cards, and PDF export in a Next.js 14 + TypeScript UI.
- Fine-tuned Mistral-7B-Instruct with QLoRA on 500+ synthetic conversation → SOAP note pairs; adapter published on Hugging Face.

Sentinel — Satellite Imagery Damage Assessment Jan – Mar 2025
PyTorch · OpenCV · Docker | github.com/ARCHIT3024/Sentinel-Damage-Assessment

- Built a two-stage PyTorch pipeline (U-Net segmentation → Siamese CNN) on 22,069+ pre/post-disaster satellite image pairs (xBD) for 5-class damage assessment; F1 0.88 (No Damage) and 0.82 (Destroyed), with Dockerized inference.

CERTIFICATIONS & ACHIEVEMENTS

- The Complete Full-Stack Web Development Bootcamp** | Udemy (Dr. Angela Yu), 62 hours | Sep 2026
- Networking Basics** | Cisco Networking Academy | Jan 2026
- AWS Certified Cloud Practitioner (CLF-C02)** | In progress
- Competitive Programming:** HackerRank 5-Star Python | CodeChef Bronze Problem Solver
- Hackathons:** iQOO City Battle Chennai 2026 (ShelfSense, mobile app) | Smart India Hackathon 2025 (Clean & Green Technology)